

B.Tech Examination-2023
Electronics and Communication Engineering
(Even Semester Regular and Supplementary)

Course Name: Principle Of Communication System, Course Code: ECEUGOE02

Time: 3.00 Hrs

Full Marks: 80

- Answer Q.1 and seven questions from rest of the questions.
- Answer all parts of a question in same place.
- Symbols have their usual meaning.

10x1=10

Choosing the correct alternative from the given options answer any ten questions.

- (a) The function of the input transducer in a communication system is
 (i) to transmit the message signal (ii) to modulate the message signal (iii) to convert message sound signal into electrical signal (iv) none of the above
- (b) Single sideband system needs
 (i) more bandwidth (ii) high power (iii) complex receiver circuit as compared to the other type of system (iv) none of these.
- (c) A 1000 KHz carrier is simultaneously amplitude modulated with 300 Hz and 2 KHz audio sine wave. The frequency which will not present in the output is
 (i) 998 KHz (ii) 999.7 KHz (iii) 1000.3 KHz (iv) 700 KHz
- (d) In commercial FM broadcasting the maximum frequency deviation is normally
 (i) 5 KHz (ii) 15 KHz (iii) 75 KHz (iv) 200 KHz
- (e) The process of transmitting two or more information signals simultaneously over the same channel is called
 (i) multiplexing (ii) telemetry (iii) detection (iv) modulation
- (f) Pre-emphasis in FM systems involves
 (i) compression of modulating signal (ii) expansion of modulating signal (iii) amplification of lower frequency components of the modulating signal (iv) amplification of higher frequency components of the modulating signal
- (g) If a signal band-limited to f_m Hz is sampled at a rate less than $2f_m$, the reconstructed signal will be
 (i) higher in magnitude (ii) smaller in magnitude (iii) have higher frequency suppressed (iv) distorted.
- (h) The non-uniform quantization leads to
 (i) improve signal to noise ratio for low level input signal (ii) reduce the probability of error (iii) reduce quantizing noise (iv) to increase signal strength.
- (i) Which is not a digital modulation system?
 (i) PAM (ii) PCM (iii) DM (iv) ADM
- (j) In PCM, if the transmission path is very long
 (i) repeater station is used (ii) pulse width may be increased (iii) pulse amplitude is increased (iv) pulse spacing is reduced
- (k) The modulation method that represents bits as different phase shifts of a carrier is known as
 (i) ASK (ii) FSK (iii) PSK (iv) MSK
- (l) The communication medium causes the signal to be
 (i) amplified (ii) modulated (iii) attenuated (iv) none of the these

2. (a) Draw the block diagram of a communication system and briefly explain the function of each block. 5
 (b) Mention different types of modulation. 2
 (c) Why do we need modulation in communication? 3
3. (a) What is meant by the term amplitude modulation? 2
 (b) Define modulation index for AM wave. 2
 (c) Derive an expression for single-tone amplitude modulated wave. 4
 (d) Write the condition for over modulated AM wave and draw the waveform of over modulated AM wave. 2
4. (a) Prove that the balanced modulator produces an output signal consisting of two side band only with the carrier removed. 5
 (b) How is a SSB-SC signal demodulated? 3
 (c) Calculate the percentage of power savings for DSB-SC signal for the percentage modulation of (i) 100% (ii) 50% 2
5. (a) What is Carson's rule? 2

- (b) Explain the generation of narrowband FM signals with suitable block diagram.
- (c) A single tone frequency modulated wave is denoted by following expression $s(t) = 2 \cos[2\pi \times 10^6 t + 10 \sin(2\pi \times 10^3 t)]$. Determine carrier frequency, modulating frequency, modulation index and maximum frequency deviation.
6. (a) Explain the varactor diode method for generation of FM signal.
- (b) Explain Pre-Emphasis in FM signal generation.
- (c) Compare between AM and FM
7. (a) State sampling theorem.
- (b) What is aliasing? How is it prevented?
- (c) Explain flat top sampling technique.
- (d) Find the Nyquist rate and the Nyquist interval for the signal $x(t) = \cos(4000\pi t) \cos(1000\pi t)$
8. (a) How is an analog signal converted to a digital signal in PCM system?
- (b) Derive the expression for transmission bandwidth in a PCM system.
- (c) A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that number of quantization level is 512. Determine transmission bandwidth and output signal to quantization noise ratio.
9. (a) Explain Delta modulation in detail with suitable block diagram.
- (b) What are slope overload distortion and granular noise in delta modulation and how it is removed in ADM?
- (c) Given a sine wave of frequency f_m and amplitude A_m applied to a delta modulator having a step size Δ . Show that slope overload distortion will occur if $A_m > \frac{\Delta}{2\pi f_m T_s}$. Here T_s is the sampling frequency.
10. (a) The binary data 10101001 is transmitted over a baseband channel. Draw the waveform for the transmitted data using following format – (i) Polar RZ (ii) Split phase Manchester Coding. (iii) AMI
- (b) What do you mean by coherent modulation technique and non-coherent modulation technique?
- (c) How is FSK signal detected?
11. (a) What do you mean by entropy?
- (b) Define lossless channel and deterministic channel with channel matrix and channel diagram.
- (c) A discrete memory less source (DMS) X has five possible outcomes with probabilities $P_1 = \frac{1}{2}, P_2 = \frac{1}{4}, P_3 = \frac{1}{8}, P_4 = \frac{1}{16}, P_5 = \frac{1}{16}$. Find entropy and information rate if there are 16 outcomes per second.

$$\cos(A+B) + \cos(A-B)$$

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